

# Kayoon Kim

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| <b>RESEARCH VISION</b>     | My research vision is to make data and algorithms serve humanity by mediating between people and AI systems. My research has been focused on (1) exposing social problems and developing data-driven solutions, (2) enhancing human decision-making while preventing marginalization.<br><b>Research Interest:</b> Human-AI Interaction, Human-Vehicle Interaction, Computational Social Science |                   |
| <b>EDUCATION</b>           | <b>University of Tübingen, Tübingen, Germany</b>                                                                                                                                                                                                                                                                                                                                                 | 2023 Oct-2026 Mar |
|                            | M.S., Quantitative Data Science, <i>Magna cum laude</i><br>Advisors: Jan Henry Belz, Prof. Augustin Kelava<br>Examiners: Dr. Tobias Grosse-Puppendahl, Prof. Holger Brandt<br>Thesis: Carmony: Multimodal Context-Aware Device Orchestration in Multi-User Automotive Environments (Tentative)                                                                                                   |                   |
|                            | <b>Chung-Ang University, Seoul, South Korea</b>                                                                                                                                                                                                                                                                                                                                                  | 2017 Mar-2023 Feb |
|                            | B.A. B.E., Sociology (Major), Computer Science (Concentration), <i>Cum laude</i><br>Advisor: Prof. Chan S. Suh<br>Thesis: Claiming for Rights: The Changing Landscape of Issues and Claims in the National Petition of South Korea, 2017-2020                                                                                                                                                    |                   |
| <b>HONORS &amp; GRANTS</b> | <b>DAAD STIBET Grant</b> , German Academic Exchange Service (DAAD)                                                                                                                                                                                                                                                                                                                               | 2025              |
|                            | <b>Research Grant</b> , Porsche AG                                                                                                                                                                                                                                                                                                                                                               | 2025              |
|                            | <b>Research Grant</b> , National Research Foundation of Korea                                                                                                                                                                                                                                                                                                                                    | 2024              |
|                            | <b>Merit-and-need-based Scholarship</b> , Seoul Scholarship Foundation                                                                                                                                                                                                                                                                                                                           | 2022              |
|                            | <b>Merit-and-need-based Scholarship</b> , Lotte Scholarship Foundation                                                                                                                                                                                                                                                                                                                           | 2021              |
|                            | <b>Community Award</b> , Ministry of the Interior and Safety, South Korea                                                                                                                                                                                                                                                                                                                        | 2020              |
|                            | Awarded at the hackathon sponsored by the Ministry of Interior and Safety of Korea for the COVID-19 project<br><b>National Scholarship (10 semesters)</b> , Korea Student Aid Foundation                                                                                                                                                                                                         |                   |
| <b>RESEARCH EXPERIENCE</b> | <b>Master Thesis Student</b> , Porsche AG Human-Centered AI Research                                                                                                                                                                                                                                                                                                                             | 2025              |
|                            | • Advisor: Jan Henry Belz<br>Implementing a context-aware system using AI agents for decision-making on optimal vehicle configurations and seamless human-vehicle interaction for all occupants                                                                                                                                                                                                  |                   |
|                            | <b>Research Assistant</b> , Methods Center, University of Tübingen                                                                                                                                                                                                                                                                                                                               | 2023-2024         |
|                            | • Advisor: Prof. Augustin Kelava<br>Developed the model to predict student dropouts in STEM classes using multivariate time-series analysis (i.e., Kalman Filter) to strengthen the educational environment                                                                                                                                                                                      |                   |
|                            | <b>Research Assistant</b> , <a href="#">Data Science Lab</a> , Chung-Ang University                                                                                                                                                                                                                                                                                                              | 2021 Summer       |
|                            | • Advisor: Prof. Yoonsik Cho<br>Implemented and evaluated recommendation systems enhancing human decision-making                                                                                                                                                                                                                                                                                 |                   |

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|                        | <b>Research Assistant, HIKE Lab:</b> Team Cayley, Chung-Ang University 2020-2021<br><ul style="list-style-type: none"> <li>• Advisor: Prof. Haklae Kim</li> <li>COVID-19: <b>Our Memory:</b> Investigated pandemic-driven social inequalities through statistical correlation analysis, text mining, and news media archiving</li> <li>COVID-19 Data Analysis A to Z: Assessed policy efficacy (i.e., pandemic screening facilities, hospital bed capacity), enhanced data visualization accessibility for the general public, and developed recommendations for public health data infrastructure improvements</li> <li>As a first case of a non-governmental digital archive, projects were <a href="#">donated to the National Library of Korea</a></li> </ul>                                                                        |
|                        | <b>Research Assistant, Da Move Lab,</b> Chung-Ang University 2020-2021<br><ul style="list-style-type: none"> <li>• Advisor: Prof. Chan S. Suh</li> <li>Conducted an analysis of South Korea's online petition system, revealing unresolved social issues; presented at the International Postgraduate and Academic Conference 2021</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>PUBLICATIONS</b>    | <b>[1] <a href="#">Mobilizing grievances in the internet age: The case of national online petitioning in South Korea, 2017–2022</a></b><br><b>Kayoon Kim,</b> Chan S. Suh<br>PLoS ONE 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>WORK EXPERIENCE</b> | <b>Research Intern,</b> Bosch Center for Artificial Intelligence (BCAI), Germany 2024<br><ul style="list-style-type: none"> <li>• Advisor: Lukas Grossberger</li> <li>Developed and tested Preferential Bayesian Optimization algorithms to efficiently optimize machinery settings for expert operators, modeling human preferences in decision-making processes and identifying cognitive biases for future human-in-the-loop experiments</li> </ul> <b>Data Quality Manager Intern,</b> Kakao Style, South Korea 2023<br><ul style="list-style-type: none"> <li>• Analyzed model prediction data and developed dashboards to enhance model accuracy and user experience. Designed and implemented an automatic labeling model to streamline the workflow</li> </ul>                                                                   |
| <b>SERVICE</b>         | Volunteer, <a href="#">Tübingen Women in Machine Learning</a> 2025<br>Volunteer, <a href="#">Soapbox Science Tübingen</a> 2025<br><ul style="list-style-type: none"> <li>• Created engaging digital content to advertise the events and connect the public with leading women scientists</li> </ul> Guest Speaker, Chung-Ang University 2022<br><ul style="list-style-type: none"> <li>• Delivered a single session lecture on the overview of computational social science and data science for an Introduction to Sociology course to engage freshmen</li> </ul> Academic Advisor, <a href="#">COSADAMA</a> 2020-2021<br><ul style="list-style-type: none"> <li>• Designed instructional sessions on Git, GitHub, and Python programming for non-STEM major students; mentored participants throughout the learning process</li> </ul> |
| <b>SKILLS</b>          | <b>Language</b><br>English (Full professional proficiency), Korean (Native)<br><b>Programming and Development</b><br>Python, R, C, SPSS, MySQL, Pytorch, Git, Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |